

UNEMPLOYMENT RATE

during 2019 - 2021

PRESENTATION BY BA PHAM

MAY 2026

INTRODUCTION

None of us will forget the COVID-19 pandemic. The world was in a panic, many people died, and the global economy stopped. Between 2019 and 2021, the world and the U.S. economy changed a lot.



MY HYPOTHESIS



Between 2019 and 2021, the unemployment rate in tourism-dependent states, such as Nevada and Hawaii, fluctuated significantly more than that in tech-driven states, such as California and Washington

DATASET

Collecting

Collecting from usda.org

Link:

<https://www.ers.usda.gov/data-products/atlas-of-rural-and-small-town-america>

Cleaning

Focusing on

Tourism-dependent states:

- Nevada
- Hawaii

Tech-driven states

- California
- Washington

COMMAND LINE

```
awk -F, '{print $2, $3, $58, $59, $41, $43, $51, $53}' Jobs_data.csv > unemploy_1
```

```
hoang@Harry-Bapham MINGW64 ~/Documents/HW1_DataIntensiveClass/HW1_data_intensive_cl
$ awk -F, '{print $2, $3, $58, $59, $41, $43, $51, $53}' Jobs_data.csv > unemploy_1
hoang@Harry-Bapham MINGW64 ~/Documents/HW1_DataIntensiveClass/HW1_data_intensive_cl
$ head -5 unemploy_labor.txt
State County NumUnemployed2019 NumCivLaborforce2019 NumUnemployed2020 NumCivLaborFo
US United States 6009990 163815890 13023736 161207103 8658650 161758337
AL Alabama 72024 2271892 147160 2268309 77272 2246993
AL Autauga 764 26684 1420 26405 742 26341
AL Baldwin 2877 98921 6159 98910 2946 99427
```

Field	Column number
State	2
County	3
NumUnemployed2019	58
NumUnemployed2020	41
NumUnemployed2021	51
NumCivLaborforce2019	59
NumCivLaborforce2020	43
NumCivLaborforce2021	53

COMMAND LINE

CALCULATING UNEMPLOYMENT RATE (%)



Unemployed

Civilian Labor Force

$\times 100 =$

**Unemployment
Rate**



Number of people
without jobs, actively
seeking work



Total number of employed
+ unemployed people

The percentage of the labor
force that is unemployed

COMMAND LINE

```
awk 'NR==1 {print}
```

```
> $1 == "HI" && $2 == "Hawaii" {print}
```

```
> $1 == "NV" && $2 == "Nevada" {print}
```

```
> $1 == "CA" && $2 == "California" {print}
```

```
> $1 == "WA" && $2 == "Washington" {print}
```

```
> ' Unemploy_labor.txt > 4_states.txt
```

COMMAND LINE

State	UnemployRate2019	UnemployRate2020	UnemployRate2021
CA	4.11%	10.22%	7.30%
HI	2.45%	12.00%	5.72%
HI	3.13%	11.67%	5.51%
NV	4.02%	13.52%	7.23%
WA	4.30%	8.47%	5.23%

SPARK

Select necessary columns

```
df = df.select(df['State'], df['County'], df['NumUnemployed2019'],df['NumUnemployed2020'],df['NumUnemployed2021'],df['NumCivLabor  
df.show()
```

-----+						
-----+						
State	County	NumUnemployed2019	NumUnemployed2020	NumUnemployed2021	NumCivLaborforce2019	NumCivLaborforce2020
rforce2021						
-----+						
-----+						
US	United States	6009990	13023736	8658650	163815890	161207103
161758337						
AL	Alabama	72024	147160	77272	2271892	2268309
2246993						
AL	Autauga	764	1420	742	26684	26405
26341						
AL	Baldwin	2877	6159	2946	98921	98910
99427						



Filter rows that consist of such states as Nevada, Hawaii, California, and Washington

Filter necessary rows

```
from pyspark.sql import functions as F
df_states = df.filter(
    ((F.col('State') == 'NV') & (F.col('County') == 'Nevada')) |
    ((F.col('State') == 'HI') & (F.col('County') == 'Hawaii')) |
    ((F.col('State') == 'CA') & (F.col('County') == 'California')) |
    ((F.col('State') == 'WA') & (F.col('County') == 'Washington'))
)
df_states.show()
```

```
+-----+-----+-----+-----+-----+-----+
|State|County|NumUnemployed2019|NumUnemployed2020|NumUnemployed2021|NumCivLaborforce2019|NumCivLaborforce2021|
+-----+-----+-----+-----+-----+-----+
|CA|California|796806|1934450|1381250|19409413|19409413|
|HI|Hawaii|16776|79512|38226|684690|684690|
```

Create UnemploymentRate column = NumUnemployed column / NumCivLaborforce column

```
from pyspark.sql import functions as F
df_div = df_states.withColumns({
    'UnemployedRate2019': F.round(F.col('NumUnemployed2019') / F.col('NumCivLaborforce2019') * 100, 2),
    'UnemployedRate2020': F.round(F.col('NumUnemployed2020') / F.col('NumCivLaborforce2020') * 100, 2),
    'UnemployedRate2021': F.round(F.col('NumUnemployed2021') / F.col('NumCivLaborforce2021') * 100, 2)
})
df_div.show()
```

State	County	NumUnemployed2019	NumUnemployed2020	NumUnemployed2021	NumCivLaborforce2019	NumCivLaborforce2021	UnemployedRate2019	UnemployedRate2020	UnemployedRate2021
CA	California	796806	1934450	1381250	19409413	8923194	4.11	10.22	7.3
HI	Hawaii	16776	79512	38226	684690	668413	2.45	12.0	5.72
NV	Nevada	61995	203208	108822	1541772				

SPARK

State County		UnemployedRate2019	UnemployedRate2020	UnemployedRate2021
CA	California	4.11	10.22	7.3
HI	Hawaii	2.45	12.0	5.72
NV	Nevada	4.02	13.52	7.23
WA	Washington	4.3	8.47	5.23

Final Table

COMPARASION

State	UnemployRate2019	UnemployRate2020	UnemployRate2021
CA	4.11%	10.22%	7.30%
HI	2.45%	12.00%	5.72%
HI	3.13%	11.67%	5.51%
NV	4.02%	13.52%	7.23%
WA	4.30%	8.47%	5.23%

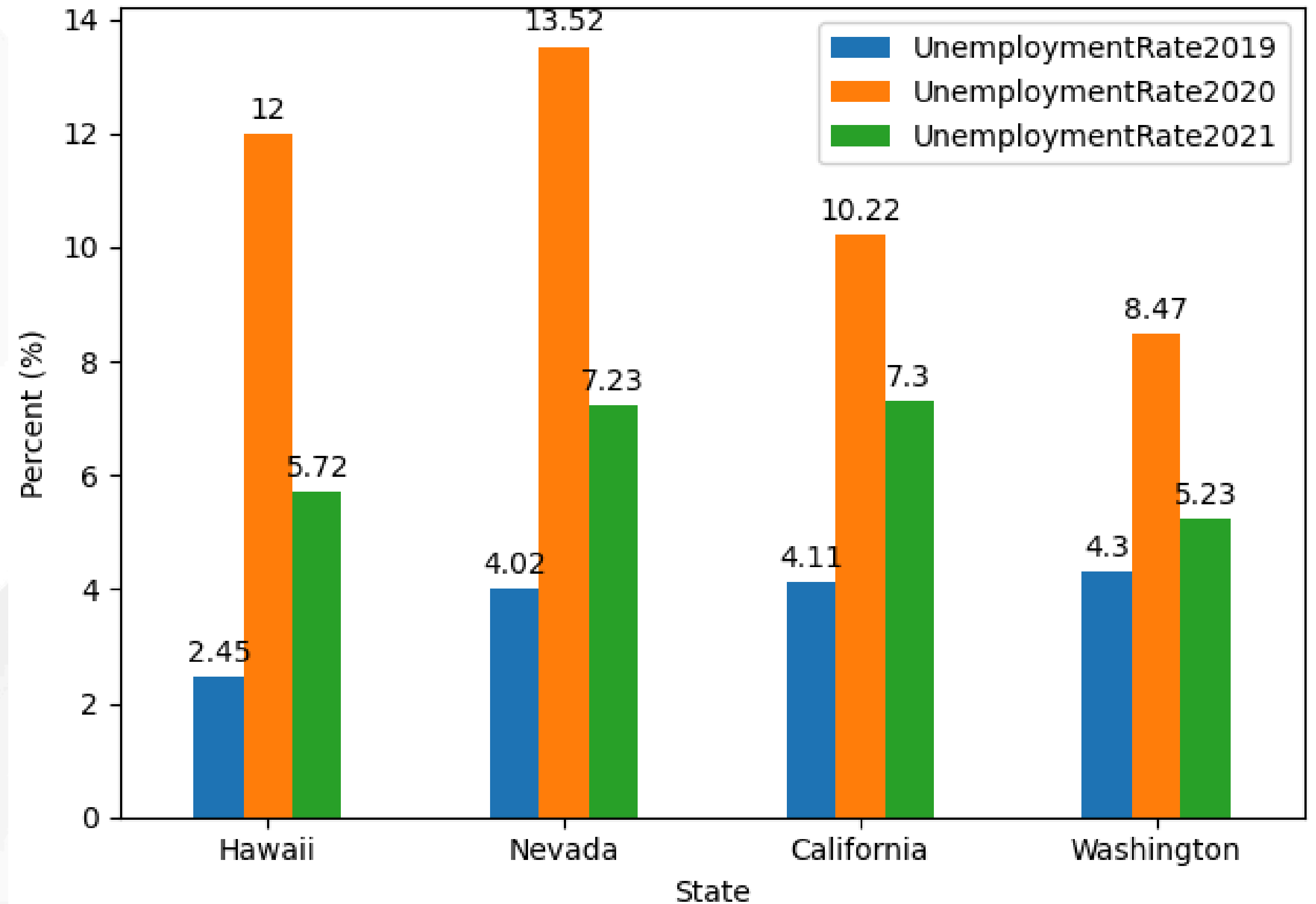
Homework 1

Homework 2

State	County	UnemployedRate2019	UnemployedRate2020	UnemployedRate2021
CA	California	4.11	10.22	7.3
HI	Hawaii	2.45	12.0	5.72
NV	Nevada	4.02	13.52	7.23
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VISUALIZATION

Unemployment Rate by State



CONCLUSION

Between 2019 and 2021, the unemployment rate in tourism-dependent states, such as Nevada and Hawaii, fluctuated significantly more than those in tech-driven states, such as California and Washington



THANK YOU